DevOps CI/CD Pipeline using Jenkins, Docker, and Kubernetes

Project Objective

Design and automate a robust CI/CD pipeline to deploy a multi-tier web application (frontend and backend) using:

- ➢ **Jenkins** for Continuous Integration and Continuous Deployment
- ➢ **Docker** for containerization
- ➢ **Kubernetes** for orchestration (via **Minikube**)
- ➢ **AWS EC2** (c7i-flex.large) as the host environment
- ➢ **GitHub** as the code repository

The goal is to build a reliable, production-ready DevOps workflow.

Hosting Environment

- ➢ **Cloud Platform**: AWS EC2
- ➢ **Instance Type**: c7i-flex.large (2 vCPU, 4 GB RAM)
- ➢ **Operating System**: Ubuntu 24.04 LTS
- ➢ **Security Group Ports Opened**: 22 (SSH), 8080 (Jenkins), 30080 (Backend), 30081 (Frontend)

➢ **Minikube Tunnel**: Enabled using minikube tunnel --bind-address 0.0.0.0

Tools & Technologies

Tool	Purpose
GitHub	Version Control
Docker	Containerize frontend/backend apps
Jenkins	Automate CI/CD pipeline
Kubernetes	Manage containerized workloads
Minikube	Local Kubernetes cluster
Nginx	Frontend static file server
Flask	Python backend web server

Project Structure

DevOps-CI-CD-Pipeline-Using-Jenkins-Docker-and

-Kubernetes/

```
├── backend/
│   ├── Dockerfile
│   ├── app.py
│   └── requirements.txt
├── frontend/
│   ├── Dockerfile
│   └── index.html
├── k8s/
│   ├── backend-deployment.yaml
│   ├── frontend-deployment.yaml
│   └── service.yaml
├── Jenkinsfile
├── docker-compose.yml
└── README.md
```

Setup and Installation

Install Required Tools

→sudo apt update && sudo apt install -y git curl docker.io

→sudo usermod -aG docker $USER

Install Minikube:

→curl -LO
https://storage.googleapis.com/minikube/releases/latest/minikube- linux-amd64

→sudo install minikube-linux-amd64
/usr/local/bin/minikube

Install kubectl:

→curl -LO "https://dl.k8s.io/release/$(curl -L -s https://dl.k8s.io/release/stable.txt)/bin/linux/amd64/kubectl"

→chmod +x kubectl && sudo mv kubectl /usr/local/bin/

Start Minikube:

→minikube start --driver=docker

→minikube tunnel --bind-address 0.0.0.0

Docker: Containerization

Backend Dockerfile

```
FROM python:3.9-slim
WORKDIR /app
COPY requirements.txt .
RUN pip install Flask==2.0.3 Werkzeug==2.0.3
COPY app.py .
EXPOSE 5000
CMD ["python", "app.py"]
```

Frontend Dockerfile

```
FROM nginx:alpine
COPY index.html /usr/share/nginx/html
EXPOSE 80
CMD ["nginx", "-g", "daemon off;"]
```

Build & Push Images

→docker build -t shivakumar46/devops-backend:latest ./backend
→docker build -t shivakumar46/devops-frontend:latest ./frontend

→docker push shivakumar46/devops-backend:latest
→docker push shivakumar46/devops-frontend:latest

Kubernetes Deployment

Backend Deployment

- Kind: Deployment
- Name: devops-backend
- Replicas: 2
- Container Image: tpathak21/devops-backend:latest
- Container Port: 5000
- Label: app: devops-backend

Purpose: Launches two backend container replicas managed by Kubernetes

Backend Service

- Kind: Service
- Name: devops-backend-service
- Type: NodePort
- Selector: app: devops-backend
- Ports:
    - Port: 5000
    - TargetPort: 5000
    - NodePort: 30080

Purpose: Exposes backend app outside the cluster on EC2 instance's IP:30080

Frontend Deployment

- Kind: Deployment
- Name: devops-frontend
- Replicas: 2
- Container Image: shivakumar46/devops-frontend:latest
- Container Port: 80
- Label: app: devops-frontend
- Purpose: Launches two frontend container replicas

Frontend Service

- Kind: Service
- Name: devops-frontend-service
- Type: NodePort
- Selector: app: devops-frontend
- Ports:
    - Port: 80
    - TargetPort: 80
    - NodePort: 30081

Purpose: Exposes frontend app outside the cluster on EC2 instance's IP:30081

Apply Resources

➔ kubectl apply -f k8s/
➔ kubectl get pods -o wide
➔ kubectl get svc

Jenkins CI/CD Pipeline

Agent Declaration

- Type: any
- Purpose: Allows pipeline to run on any available Jenkins agent (node)

Environment Setup

- DOCKERHUB_CREDENTIALS:
- Secure Jenkins credentials (ID: dockerhub-creds) used for DockerHub login

Stages

Stage 1: Checkout Code from GitHub Clones the main branch from your GitHub repository: https://github.com/shivakumarshivas/Devops-Project.git

Stage 2: Build Backend Docker Image

- Builds Docker image:
  - Image: shivakumar46/devops-backend:latest
  - Context: ./backend

Stage 3: Build Frontend Docker Image

- Builds Docker image:
  - Image: shivakumar46/devops-frontend:latest
  - Context: ./frontend

Stage 4: Login to DockerHub

- Authenticates Jenkins with DockerHub using securely stored credentials

Stage 5: Push Docker Images

- Pushes both frontend and backend images to DockerHub:
  - shivakumar46/devops-backend:latest
  - shivakumar46/devops-frontend:latest

Stage 6: Deploy to Kubernetes

- Uses kubectl to apply manifests for:
  - Backend Deployment
  - Frontend Deployment
  - NodePort Services
- Uses the Kubeconfig file located at: /var/lib/jenkins/.kube/config

<u>End Result</u>

- Fully automated CI/CD pipeline from GitHub to DockerHub to a live Kubernetes cluster

- Runs seamlessly inside an AWS EC2 instance with Jenkins managing every stage

Output Validation

curl Test

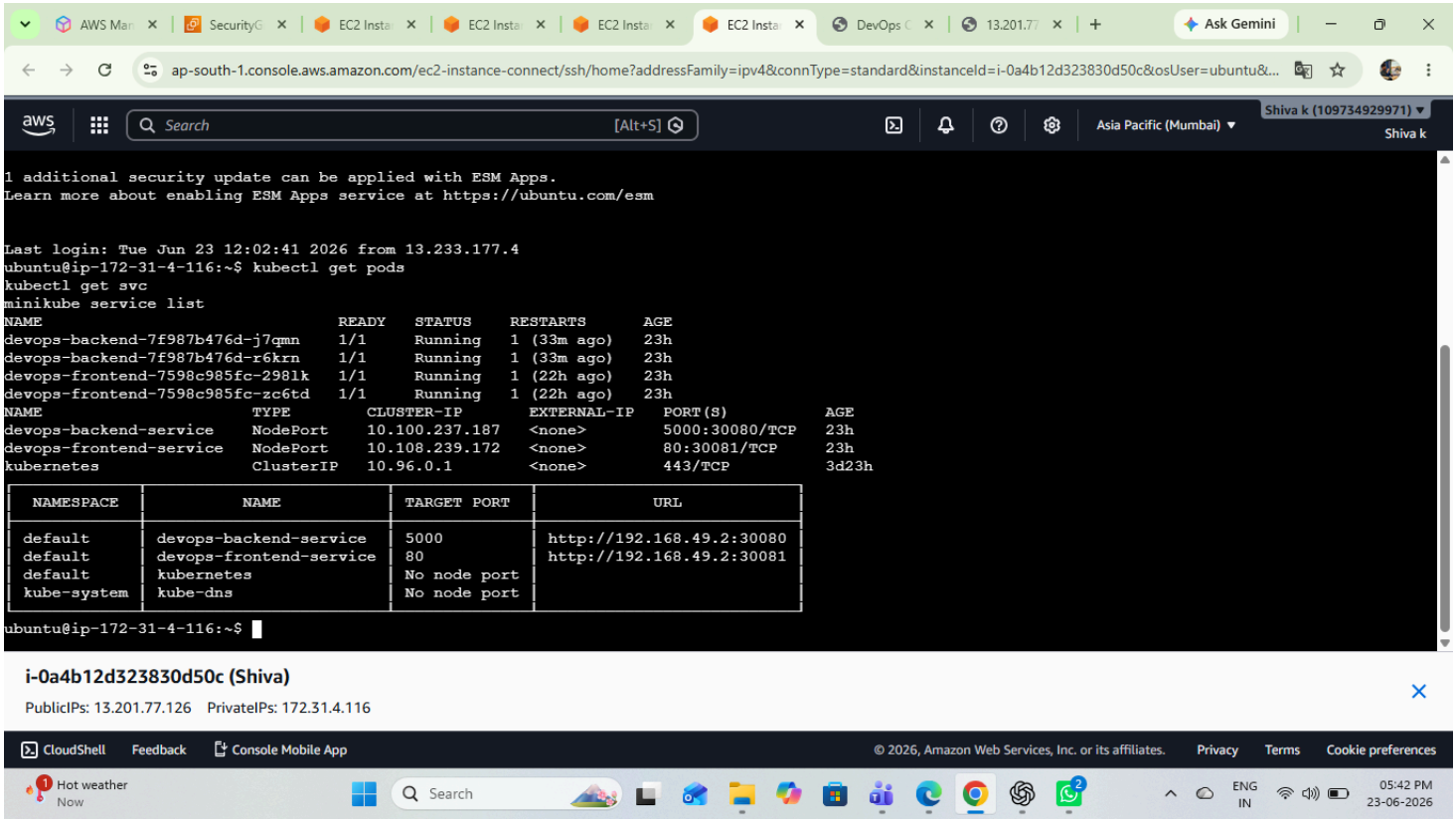
➢ curl http://<EC2-PUBLIC-IP>:30080 # Backend

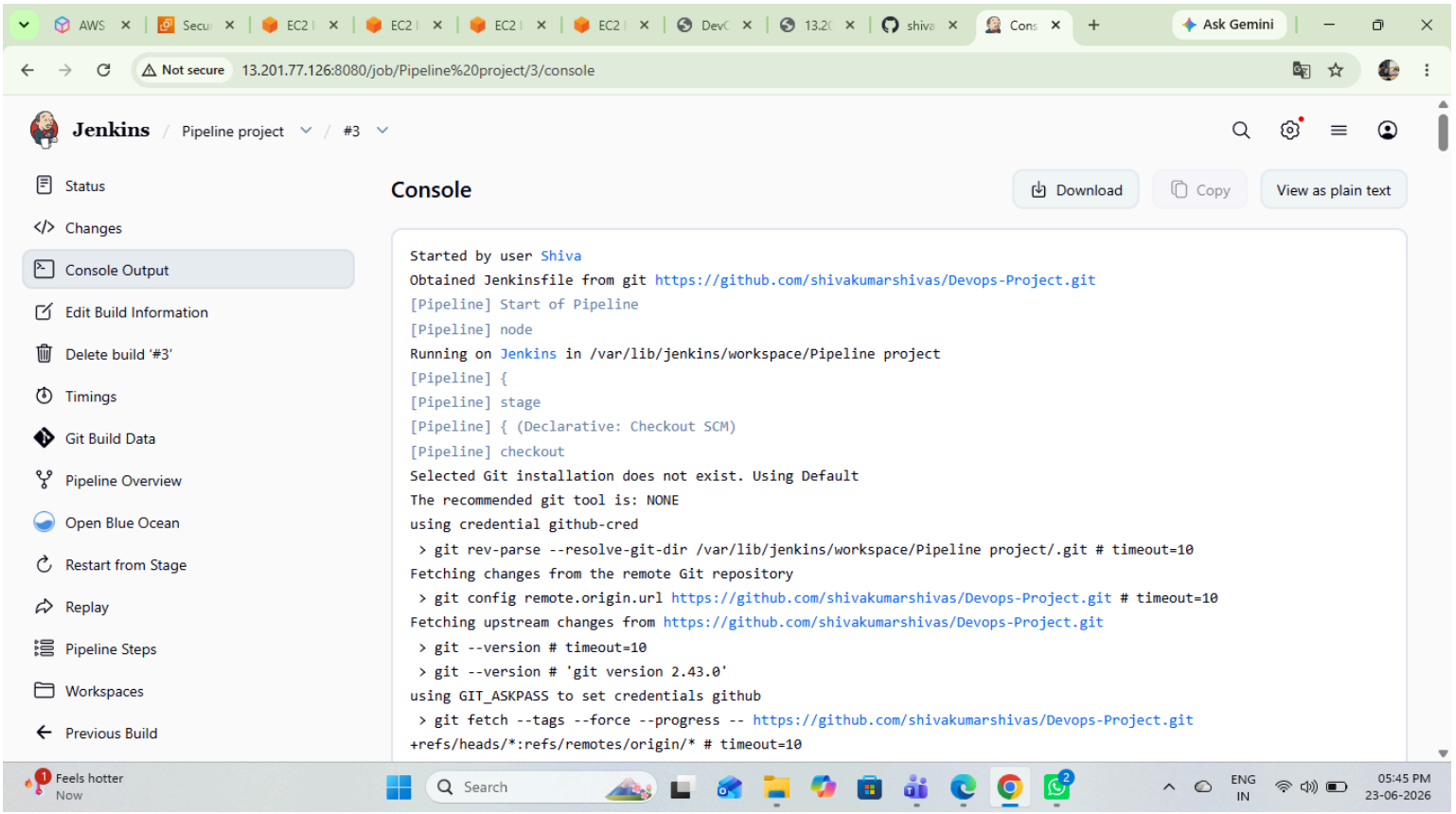
➢ curl http://<EC2-PUBLIC-IP>:30081 # Frontend

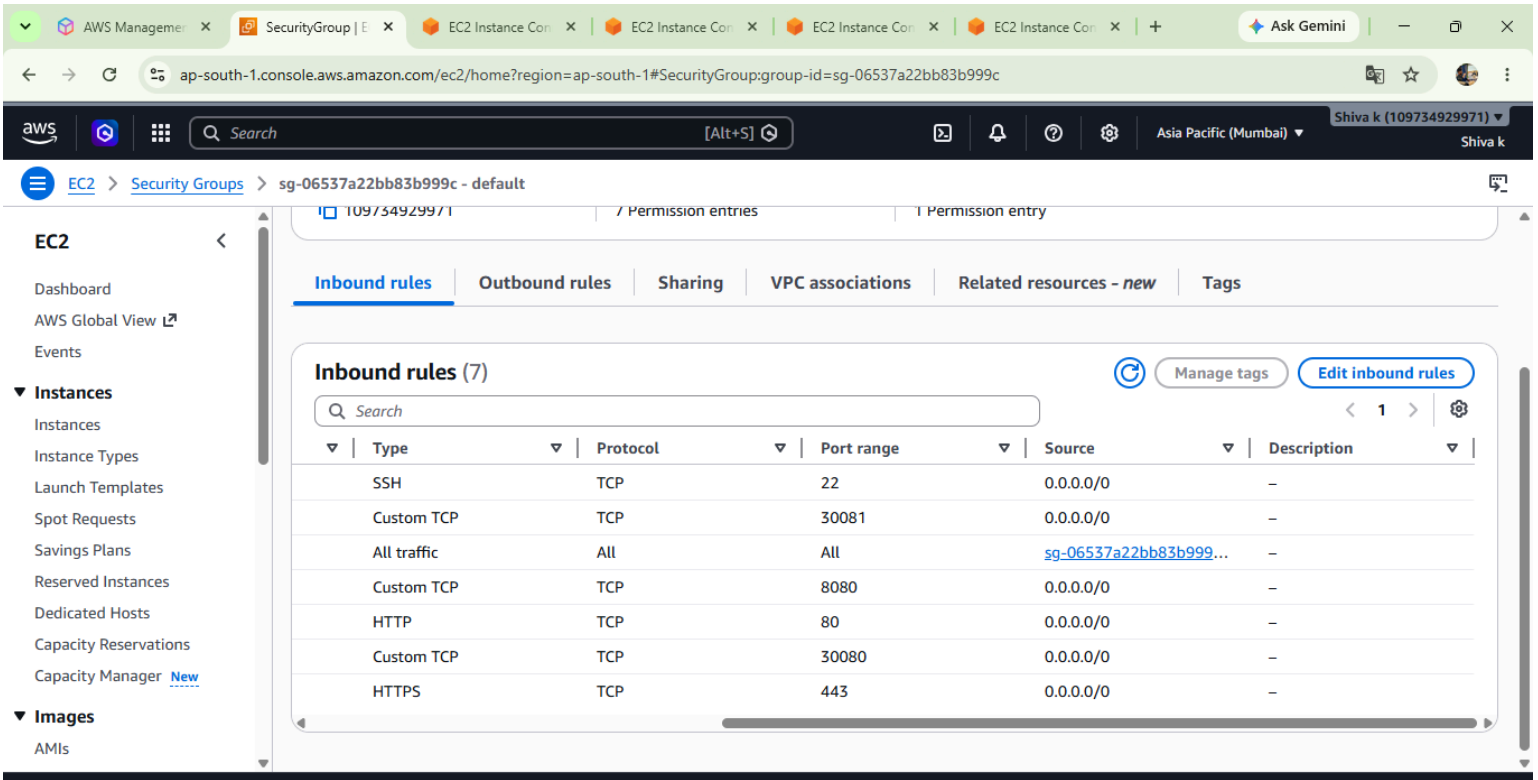
Jenkins Pipeline

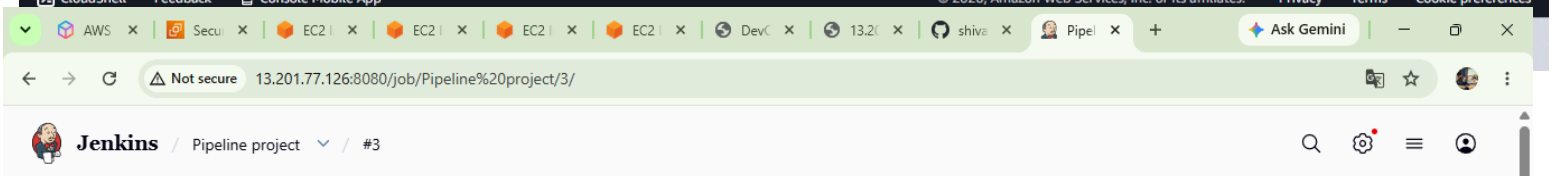
All stages: Clone > Build > Push > Deploy should show as green

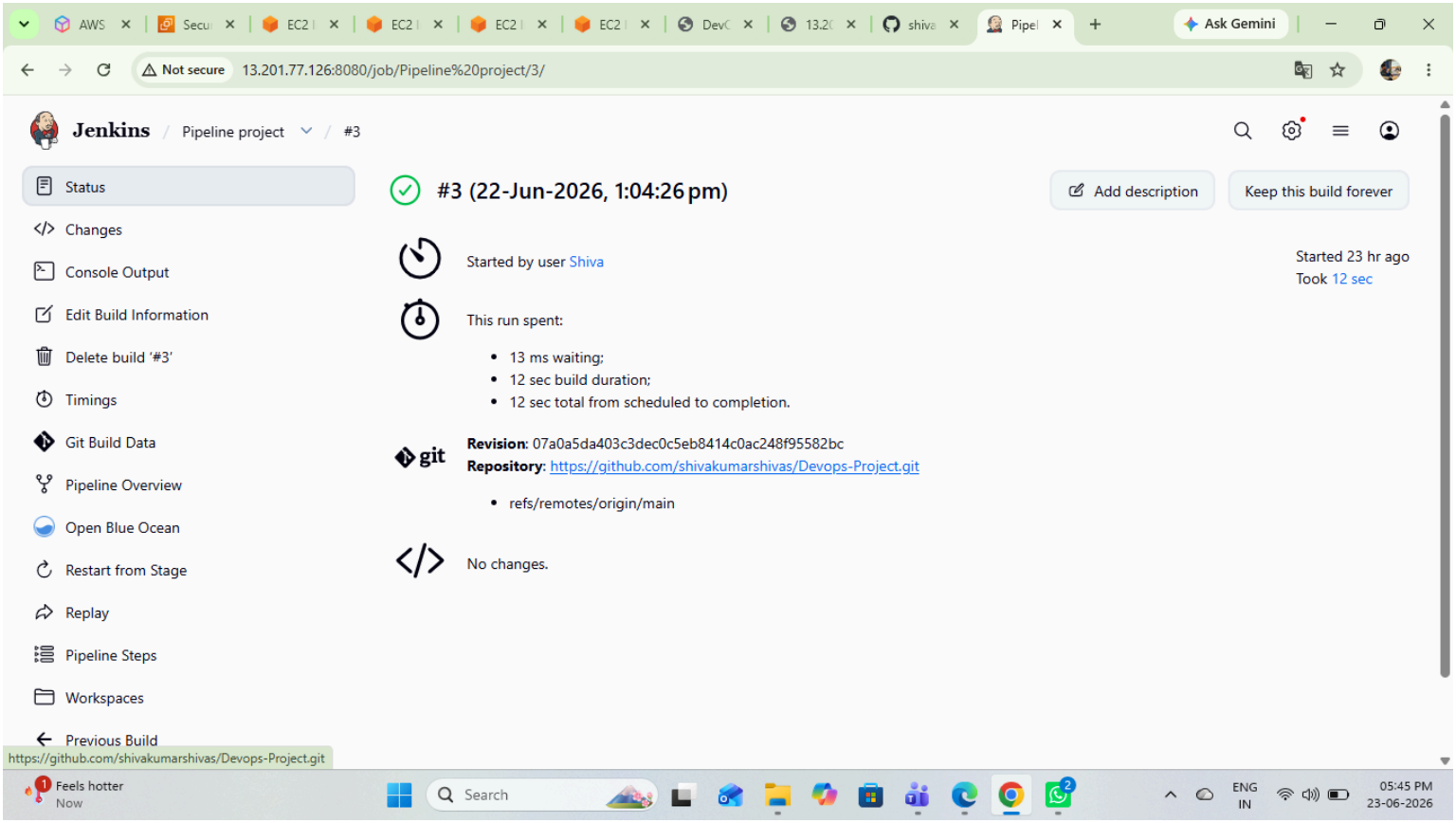
Outputs Images

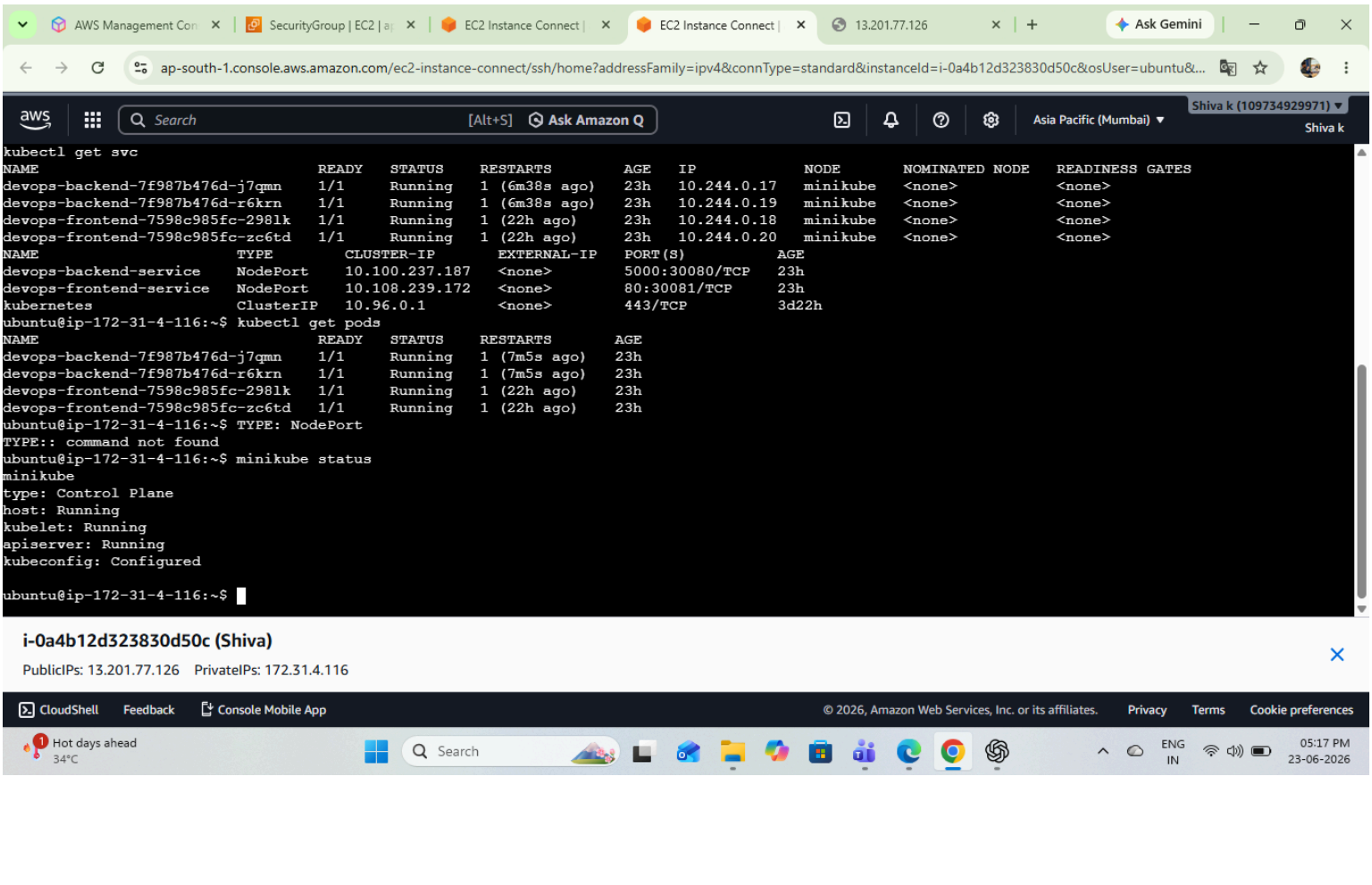


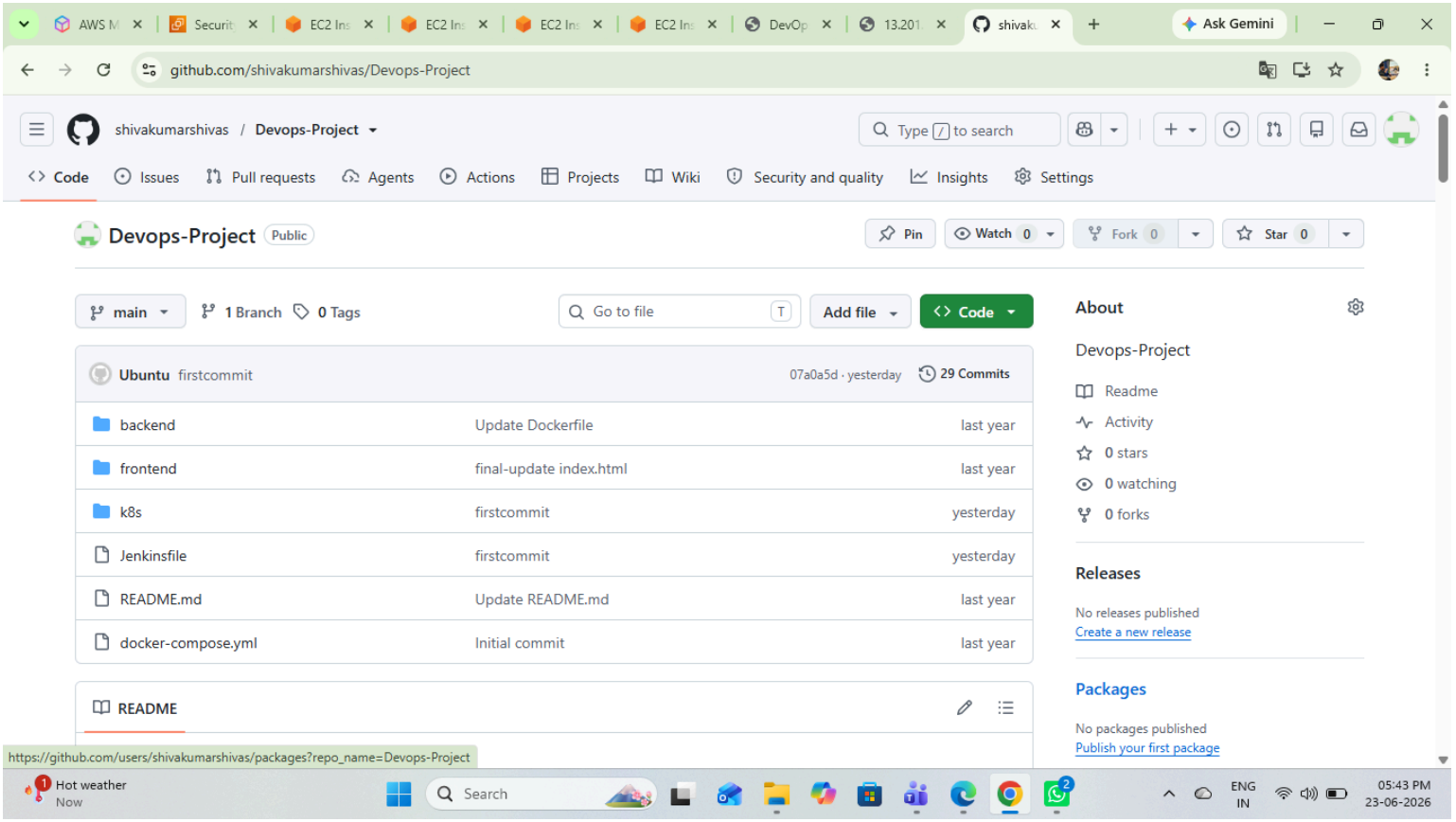

Not secure 13.201.77.126:8080/job/Pipeline%20project/3/console
Jenkins / Pipeline project / #3
Status
Changes
Console Output
Edit Build Information
Delete build '#3'
Timings
Git Build Data
Pipeline Overview
Open Blue Ocean
Restart from Stage
Replay
Pipeline Steps
Workspaces
Previous Build
Console
Download
Copy
View as plain text
Started by user Shiva
Obtained Jenkinsfile from git https://github.com/shivakumarshivas/Devops-Project.git
[Pipeline] Start of Pipeline
[Pipeline] node
Running on Jenkins in /var/lib/jenkins/workspace/Pipeline project
[Pipeline] {
[Pipeline] stage
[Pipeline] { (Declarative: Checkout SCM)
[Pipeline] checkout
Selected Git installation does not exist. Using Default
The recommended git tool is: NONE
using credential github-cred
> git rev-parse --resolve-git-dir /var/lib/jenkins/workspace/Pipeline project/.git # timeout=10
Fetching changes from the remote Git repository
> git config remote.origin.url https://github.com/shivakumarshivas/Devops-Project.git # timeout=10
Fetching upstream changes from https://github.com/shivakumarshivas/Devops-Project.git
> git --version # timeout=10
> git --version # 'git version 2.43.0'
using GIT_ASKPASS to set credentials github
> git fetch --tags --force --progress -- https://github.com/shivakumarshivas/Devops-Project.git
+refs/heads/*:refs/remotes/origin/* # timeout=10

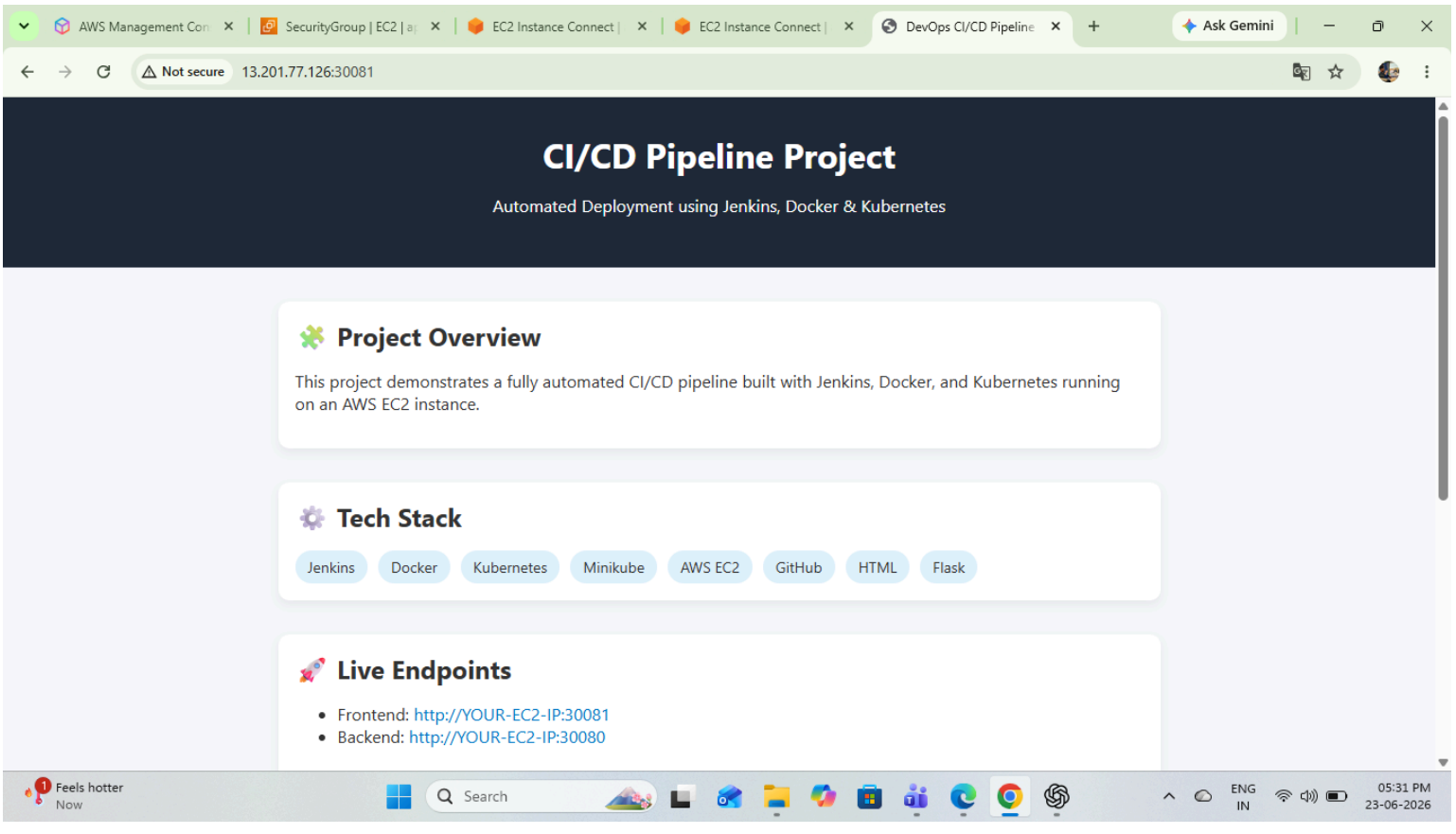

ap-south-1.console.aws.amazon.com/ec2/home?region=ap-south-1#SecurityGroup:group-id=sg-06537a22bb83b999c
Search [Alt+S]
Asia Pacific (Mumbai)
Shiva k (109734929971)
EC2 > Security Groups > sg-06537a22bb83b999c - default
EC2
Dashboard
AWS Global View
Events
Instances
Instances
Instance Types
Launch Templates
Spot Requests
Savings Plans
Reserved Instances
Dedicated Hosts
Capacity Reservations
Capacity Manager New
Images
AMIs
Inbound rules
Outbound rules
Sharing
VPC associations
Related resources - new
Tags
Inbound rules (7)
Manage tags
Edit inbound rules
Search
Type | Protocol | Port range | Source | Description
SSH | TCP | 22 | 0.0.0.0/0 | –
Custom TCP | TCP | 30081 | 0.0.0.0/0 | –
All traffic | All | All | sg-06537a22bb83b999... | –
Custom TCP | TCP | 8080 | 0.0.0.0/0 | –
HTTP | TCP | 80 | 0.0.0.0/0 | –
Custom TCP | TCP | 30080 | 0.0.0.0/0 | –
HTTPS | TCP | 443 | 0.0.0.0/0 | –
CloudShell
Feedback
Console Mobile App
© 2026, Amazon Web Services, Inc. or its affiliates.
Privacy
Terms
Cookie preferences

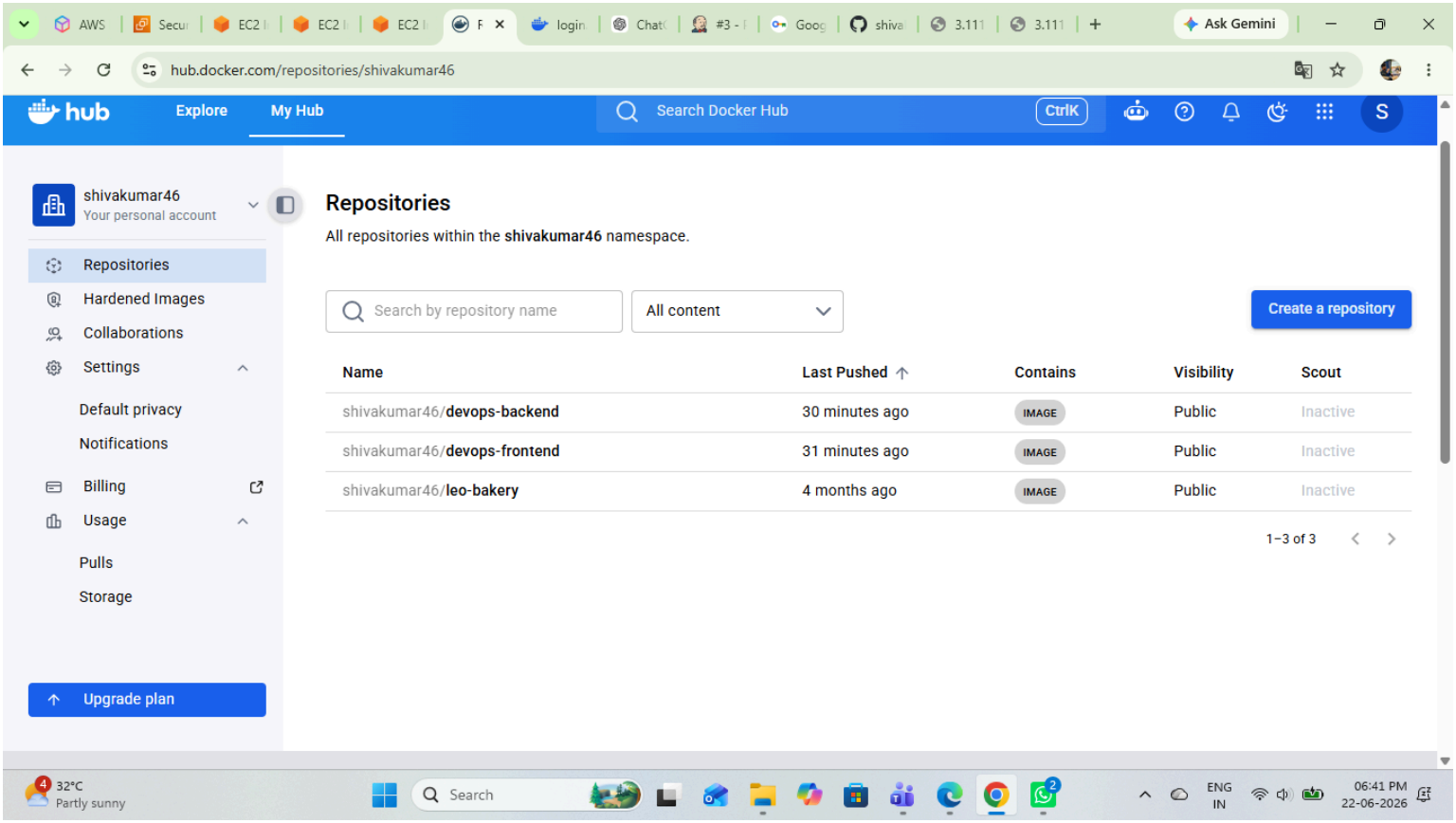

Not secure 13.201.77.126:8080/job/Pipeline%20project/3/
Jenkins / Pipeline project / #3

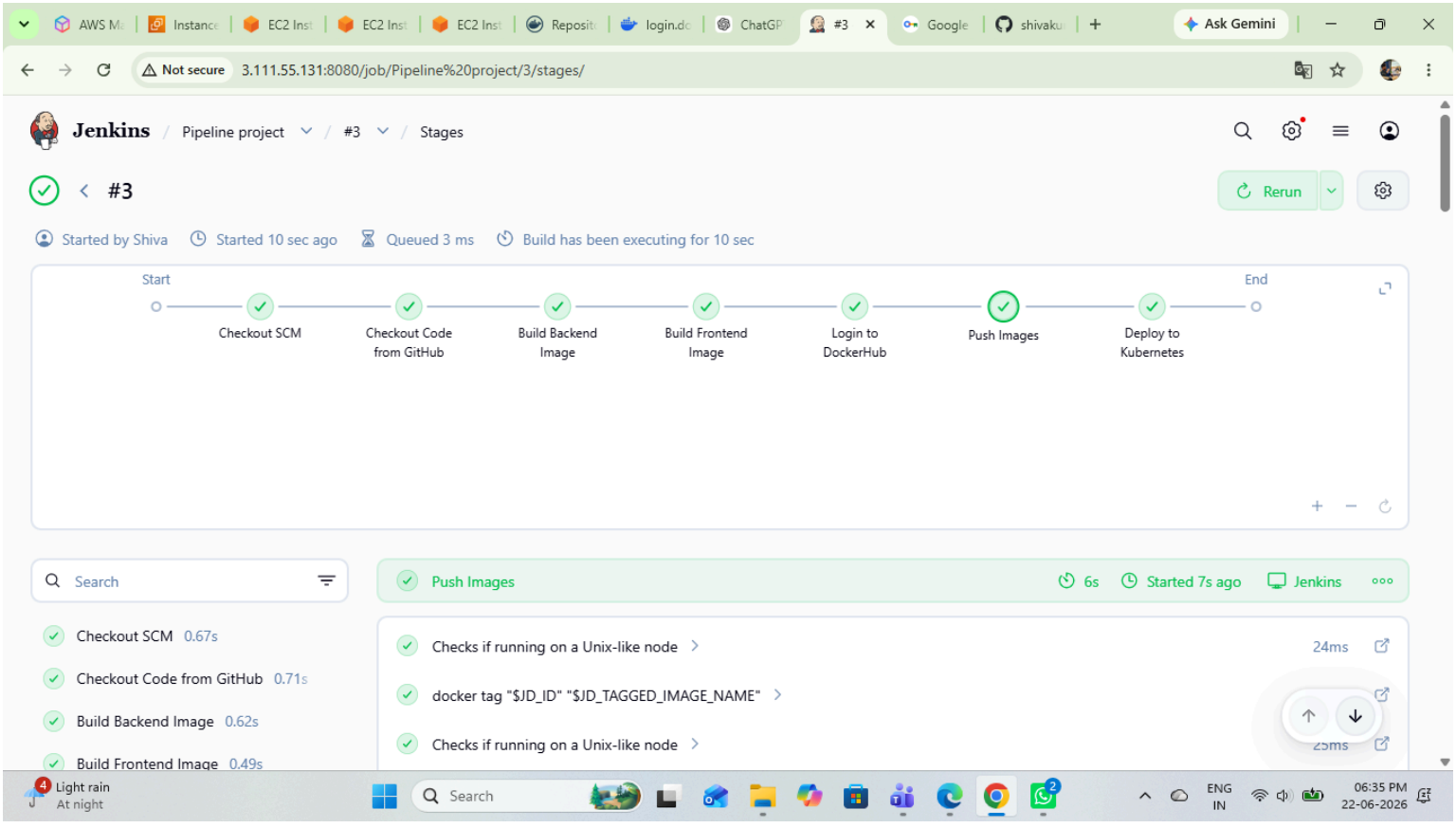

Not secure
13.201.77.126:8080/job/Pipeline%20project/3/
Jenkins / Pipeline project / #3
Status
Changes
Console Output
Edit Build Information
Delete build '#3'
Timings
Git Build Data
Pipeline Overview
Open Blue Ocean
Restart from Stage
Replay
Pipeline Steps
Workspaces
Previous Build
#3 (22-Jun-2026, 1:04:26 pm)
Add description
Keep this build forever
Started by user Shiva
Started 23 hr ago
Took 12 sec
This run spent:
13 ms waiting;
12 sec build duration;
12 sec total from scheduled to completion.
Revision: 07a0a5da403c3dec0c5eb8414c0ac248f95582bc
Repository: https://github.com/shivakumarshivas/Devops-Project.git
refs/remotes/origin/main
No changes.
https://github.com/shivakumarshivas/Devops-Project.git


ap-south-1.console.aws.amazon.com/ec2-instance-connect/ssh/home?addressFamily=ipv4&connType=standard&instanceId=i-0a4b12d323830d50c&osUser=ubuntu&...
kubectl get svc
NAME READY STATUS RESTARTS AGE IP NODE NOMINATED NODE READINESS GATES
devops-backend-7f987b476d-j7qmn 1/1 Running 1 (6m38s ago) 23h 10.244.0.17 minikube <none> <none>
devops-backend-7f987b476d-r6krn 1/1 Running 1 (6m38s ago) 23h 10.244.0.19 minikube <none> <none>
devops-frontend-7598c985fc-298lk 1/1 Running 1 (22h ago) 23h 10.244.0.18 minikube <none> <none>
devops-frontend-7598c985fc-zc6td 1/1 Running 1 (22h ago) 23h 10.244.0.20 minikube <none> <none>
NAME TYPE CLUSTER-IP EXTERNAL-IP PORT(S) AGE
devops-backend-service NodePort 10.100.237.187 <none> 5000:30080/TCP 23h
devops-frontend-service NodePort 10.108.239.172 <none> 80:30081/TCP 23h
kubernetes ClusterIP 10.96.0.1 <none> 443/TCP 3d22h
ubuntu@ip-172-31-4-116:~$ kubectl get pods
NAME READY STATUS RESTARTS AGE
devops-backend-7f987b476d-j7qmn 1/1 Running 1 (7m5s ago) 23h
devops-backend-7f987b476d-r6krn 1/1 Running 1 (7m5s ago) 23h
devops-frontend-7598c985fc-298lk 1/1 Running 1 (22h ago) 23h
devops-frontend-7598c985fc-zc6td 1/1 Running 1 (22h ago) 23h
ubuntu@ip-172-31-4-116:~$ TYPE: NodePort
TYPE:: command not found
ubuntu@ip-172-31-4-116:~$ minikube status
minikube
type: Control Plane
host: Running
kubelet: Running
apiserver: Running
kubeconfig: Configured
ubuntu@ip-172-31-4-116:~$
i-0a4b12d323830d50c (Shiva)
PublicIPs: 13.201.77.126 PrivateIPs: 172.31.4.116
CloudShell Feedback Console Mobile App
© 2026, Amazon Web Services, Inc. or its affiliates. Privacy Terms Cookie preferences


github.com/shivakumarshivas/Devops-Project
shivakumarshivas / Devops-Project
Code
Issues
Pull requests
Agents
Actions
Projects
Wiki
Security and quality
Insights
Settings
Devops-Project
Public
Pin
Watch 0
Fork 0
Star 0
main
1 Branch
0 Tags
Go to file
Add file
Code
Ubuntu firstcommit
07a0a5d · yesterday
29 Commits
backend Update Dockerfile last year
frontend final-update index.html last year
k8s firstcommit yesterday
Jenkinsfile firstcommit yesterday
README.md Update README.md last year
docker-compose.yml Initial commit last year
README
About
Devops-Project
Readme
Activity
0 stars
0 watching
0 forks
Releases
No releases published
Create a new release
Packages
No packages published
Publish your first package
https://github.com/users/shivakumarshivas/packages?repo_name=Devops-Project


13.201.77.126:30081
CI/CD Pipeline Project
Automated Deployment using Jenkins, Docker & Kubernetes
Project Overview
This project demonstrates a fully automated CI/CD pipeline built with Jenkins, Docker, and Kubernetes running on an AWS EC2 instance.
Tech Stack
Jenkins
Docker
Kubernetes
Minikube
AWS EC2
GitHub
HTML
Flask
Live Endpoints
Frontend: http://YOUR-EC2-IP:30081
Backend: http://YOUR-EC2-IP:30080


hub.docker.com/repositories/shivakumar46
hub
Explore
My Hub
Search Docker Hub
CtrlK
shivakumar46
Your personal account
Repositories
Hardened Images
Collaborations
Settings
Default privacy
Notifications
Billing
Usage
Pulls
Storage
Upgrade plan
Repositories
All repositories within the shivakumar46 namespace.
Search by repository name
All content
Create a repository
Name
Last Pushed
Contains
Visibility
Scout
shivakumar46/devops-backend
30 minutes ago
IMAGE
Public
Inactive
shivakumar46/devops-frontend
31 minutes ago
IMAGE
Public
Inactive
shivakumar46/leo-bakery
4 months ago
IMAGE
Public
Inactive
1–3 of 3


Not secure
3.111.55.131:8080/job/Pipeline%20project/3/stages/
Jenkins
Pipeline project
#3
Stages
#3
Rerun
Started by Shiva
Started 10 sec ago
Queued 3 ms
Build has been executing for 10 sec
Start
End
Checkout SCM
Checkout Code from GitHub
Build Backend Image
Build Frontend Image
Login to DockerHub
Push Images
Deploy to Kubernetes
Search
Checkout SCM 0.67s
Checkout Code from GitHub 0.71s
Build Backend Image 0.62s
Build Frontend Image 0.49s
Push Images
6s
Started 7s ago
Jenkins
Checks if running on a Unix-like node
24ms
docker tag "$JD_ID" "$JD_TAGGED_IMAGE_NAME"
Checks if running on a Unix-like node

Links

GitHub Repository:
https://github.com/shivakumarshivas/Devops-Project.git
LinkedIn Profile:
https://www.linkedin.com/in/shiva-kumar11/
DockerHub Profile:
https://hub.docker.com/repositories/shivakumar46

Conclusion

This project is a full-fledged demonstration of how to containerize applications, automate deployments using Jenkins, and orchestrate containers via Kubernetes - all deployed on an AWS EC2 instance. It proves your ability to integrate multiple DevOps tools in a production-like setup.